

Usman Ali

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WORK EXPERIENCE

- **Doctoral Researcher, University of Hamburg** Jul 2026 – Present
Conducting research on privacy-preserving federated learning and virtual twins for personalized Type 2 diabetes treatment prediction. Developing machine learning and AI methods for secure biomedical data analysis in collaborative healthcare settings.
- **Graduate Teaching Assistant, University of Glasgow (Joint UESTC)** July 2025 – July 2026
Led research projects involving interviews, data analysis, literature reviews, and ethics management. Provided course support and graded assignments.
- **Python Developer, Broadstone Technologies** Aug 2022 – Oct 2024
Developed Meezy Loyalty App, increasing user engagement by 40%. Revamped Prasl 3D AR platform, boosting transactions by 25%. Launched Cloaked Privacy VPN using WireGuard, accelerating client acquisition by 60%.
- **Automation Engineer , MLC Pakistan** Apr 2022 – Jul 2022
Built automated workflows, AI models, data pipelines, and performed statistical analyses.
- **AI Intern, Ekkel.ai** Dec 2021 – Mar 2022
Developed AI models and APIs, built data pipelines, and analyses to support data-driven decisions.

EDUCATION AND TRAINING

- **University of Electronic Science and Technology of China (UESTC)** Aug 2024 – July 2026
Master of Science in Information and Communication Engineering
Current GPA: 3.91/4.0 (89%) — Ranked in the top 2% of the program.
Thesis: Nuclei Segmentation using Deep Learning
- **Virtual University of Pakistan (VU)** Apr 2019 – Apr 2023
Bachelor of Science in Computer Science (BSCS)
Final GPA: 3.60/4.0 (80%) — Ranked in the top 5% of the program.
Project: Online Fraud Detection System using machine learning.

PUBLICATIONS

- **Multi-Class Nuclei Segmentation using AI with Adaptive Sliding Window**
Authors: Usman Ali, Jin Qi (Supervisor), Aiman Rashid
Journal: International Journal of Imaging Systems and Technology (Wiley) 2026
DoI: 10.1002/ima.70283
- **Attention-Based Segmentation Model (Submitted)**
Authors: Usman Ali, Aiman Rashid , Jin Qi (Supervisor)
Journal: Computerized Medical Imaging and Graphics(Elsevier) 2026

PROFESSIONAL CERTIFICATE

- **Foundations of Project Management, Google**
Introduction to project management principles, tools, and methodologies to effectively manage projects. Nov 2022
- **Data Analytics, Google**
Comprehensive specialization covering data preparation, exploration, visualization, and making data-driven decisions. Aug 2022

- **Artificial Intelligence Specialization, NAVTCC**

Specialization covering foundations of AI, data preparation, model building, and deployment techniques.

Sep 2020 – Mar 2021

LANGUAGE SKILLS

- **Urdu:** Native proficiency
- **Punjabi:** Native proficiency
- **English:** Proficient — B2 level in listening, reading, writing, and speaking
- **Chinese:** Basic — A1–A2 level in listening, reading, writing, and speaking

HONOURS AND AWARDS

- **China Scholarship Council (CSC)**
Fully funded 2-year Master's program scholarship awarded by the Government of China. Aug 2024 - Present
- **Fully Funded Artificial Intelligence Course, Government of Pakistan**
Government-funded scholarship for advanced AI training. Nov 2020
- **Merit-Based Scholarship (4th Semester), Virtual University of Pakistan**
Awarded based on academic excellence. Sep 2020
- **Best Attendance Award, Punjab Group of Colleges**
Recognized for excellent attendance record. Sep 2018
- **Merit-Based Full Scholarship, Punjab Group of Colleges**
Awarded for outstanding academic performance. Sep 2016

SKILLS

- **Communication & Academic:** Academic writing, peer-reviewed publication writing, research proposals, technical reports, scientific presentations, and interdisciplinary communication
- **Machine Learning & AI:** Computer Vision, Deep Learning; image segmentation, classification, and representation learning; PyTorch, TensorFlow, TensorRT, Scikit-image, OpenCV; ML algorithms including CNNs, RNNs, SVM, Decision Trees; Python scientific stack (NumPy, pandas, SciPy, matplotlib, scikit-learn)
- **Medical Image Analysis:** Medical image preprocessing, feature extraction, evaluation metrics (Dice, IoU, accuracy), dataset curation, annotation workflows, and reproducible experimentation
- **Big Data & Databases:** Data preprocessing, MongoDB, PostgreSQL, data pipelines, and statistical analysis
- **Technical:** Linux (Ubuntu, CentOS), C/C++, Python, Git, REST APIs, Flask, software debugging and integration, and deployment
- **Organizational & Management:** Team collaboration, leadership, task management, research planning, problem-solving, and project organization

CERTIFICATIONS

- **Crash Course on Python (Coursera)**
Covered Python fundamentals, scripting, automation, and problem-solving. Nov 2021
- **Introduction to TensorFlow for AI, Machine Learning, and Deep Learning**
Learned TensorFlow best practices and built neural networks for ML and CV tasks. Dec 2020
- **NDG Linux Unhatched (Cisco)**
Introduced to Linux essentials, command-line usage, and IT fundamentals. Apr 2020

CONFERENCES AND SEMINARS

- **International Online Conference on Digitalization and Cyber Security**
Virtual University of Pakistan Aug 2020
- **14th International Conference on Open Source Systems and Technologies 2022**
University of Engineering and Technology, Lahore Aug 2022